polygon, and change the constraints considering these circles instead of the actual shape of the vertices.

Using circles make our life a lot easier: it allows us to just check the distance between two vertices (between their centers) or between a vertex and an edge.

For two vertices, the algorithm is trivial: we just need to check the distance between their centers is larger than the sum of their radii. Figure 15.27 shows why.

To make sure an edge doesn't overlap with a vertex (which is not an endpoint for the edge), we need to make sure that the distance between the edge and the vertex is larger than the vertex's radius.

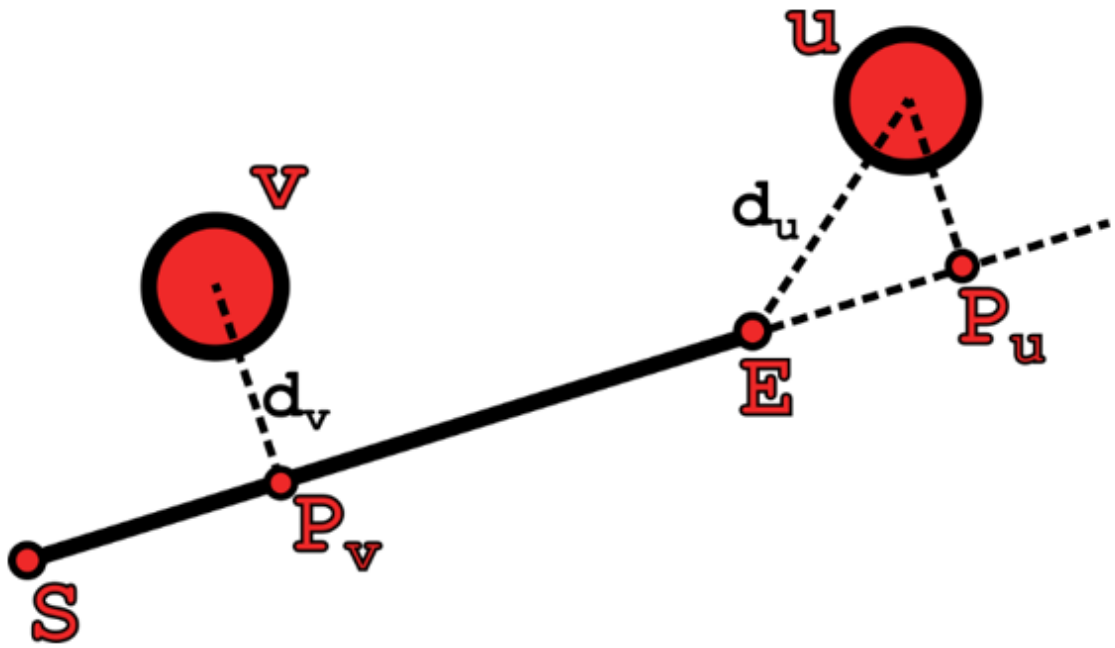


Figure 15.28 Vertex intersection with a segment. When considering the distance between a segment and a vertex, there are cases (v in the figure) where